PROBABILISTIC LINKAGE USING SPLINK

WITH APPLICATION TO CRASH OUTCOMES DATA EVALUATION SYSTEM (CODES)

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Questions for the audience
(fill out now!)

Downloadable workshop materials

divyankapawaskar.github.io/savir2026/

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PROBABILISTIC LINKAGE USING SPLINK

WITH APPLICATION TO CRASH OUTCOMES DATA EVALUATION SYSTEM (CODES)

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04.15.2026 | 2026 SAVIR Annual Conference | Providence, RI

WORKSHOP LEARNING OBJECTIVES

1. Understand the intuition behind probabilistic linkage and the Fellegi-Sunter model on a high level.
2. Understand key concepts in probabilistic linkage: blocking rules, comparisons, match weights, prediction thresholds, and how a user can fine-tune these to obtain more efficient, accurate data linkages.
3. Implement simple probabilistic linkage of two data sources using Splink.
4. Interpret output and visualizations to evaluate linkage quality.

WORKSHOP OUTLINE

Part 1: Lecture with Activity

Part 2: Software Demonstration Using Splink

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PART 1:
THE INTUITION BEHIND PROBABILISTIC LINKAGE

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BACKGROUND

CRASH OUTCOMES DATA EVALUATION SYSTEM (CODES)

Goal of CODES:

- Link the crash data with health outcomes data to measure the real injury and cost consequences of crashes.

Research questions:

- What is the injury burden of traffic crashes in your state?
- What is the financial burden of these injuries?
- How effective are safety interventions (seat belts, airbags, child restraints, motorcycle helmets, graduated licensing) at reducing injury severity, hospitalization, and death?
- What is the full injury profile of vulnerable road users (motorcyclists, pedestrians, cyclists, teen drivers, older drivers)?

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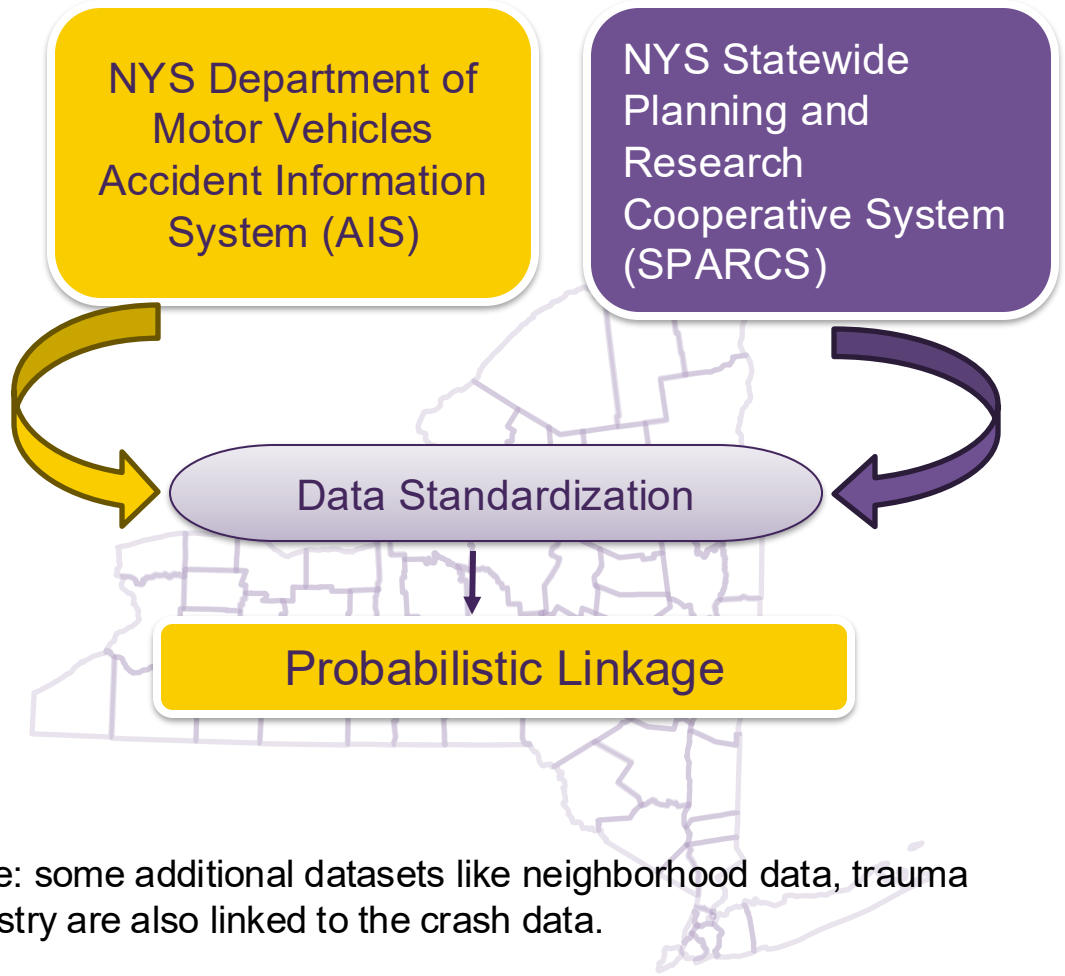
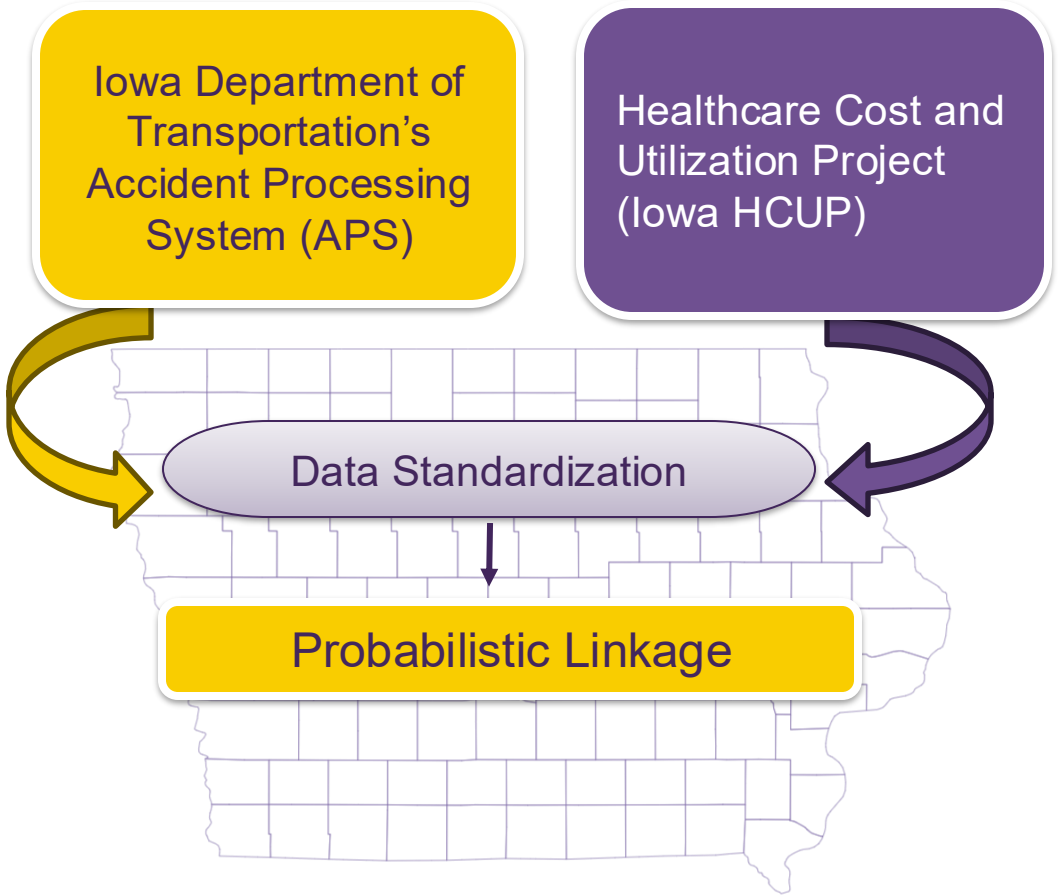
CRASH OUTCOME DATA EVALUATION SYSTEM (CODES)





Note: some additional datasets like neighborhood data, trauma registry are also linked to the crash data.

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VARIABLES WITH LINKAGE POTENTIAL (NY)

- Example: New York (higher information setting)

Crash Variables (AIS data)	Hospital Variables (SPARCS data)	
Date of crash	Admit date	Crash-related variables
Hour of crash	Admit hour	
Role *(vehicle driver/passenger, motorcycle driver/passenger, cyclist, pedestrian)*	Role *(vehicle driver/passenger, motorcycle driver/passenger, cyclist, pedestrian)*	
First Name *(first two letters)*	First Name *(first two letters)*	Demographic variables
Last Name *(first two letters)*	Last Name *(first two letters)*	
Last Name *(last two letters)*	Last Name *(last two letters)*	
Sex	Sex	
Date of birth	Date of birth	
Age	Age	
County	County	
ZIP code *(from driver's license)*	ZIP code *(from residential address)*	

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VARIABLES WITH LINKING POTENTIAL (IA)

- Example: Iowa (lower information setting)

Crash Variables (DOT's APS)	Hospital Variables (HCUP)
Date of crash	Month of admission
Birthdate	Month of birth
Sex	Sex
ZIP code *(from driver's license)*	ZIP code *(residential)*
Role in crash *(vehicle driver/passenger, motorcycle driver/passenger, cyclist, pedestrian)*	Role in crash *(vehicle driver/passenger, motorcycle driver/passenger, cyclist, pedestrian)*
Vehicle Type *(car, truck, motorcycle, etc.)*	Vehicle Type *(car, truck, motorcycle, etc.)*

Iowa HCUP does not provide full admission date or birth date; **only the month and year**

Derived from ICD-10-CM External Cause codes

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Just another day as a researcher in a perfect world…

Crash Dataset				
Unique ID	**First**	**Last**	**State**	**Crash Variables**
0123	Stephanie	Jansson	IA	…
0345	Divyanka	Pawaskar	NY	…
0567	Tara	Sangal	NY	…

Hospital dataset				
Unique ID	**First**	**Last**	**State**	**Hospital Variables**
0123	Stephanie	Jansson	IA	…
0567	Tara	Sangal	NY	…
0345	Divyanka	Pawaskar	NY	…

Linking these is easy! Just do deterministic linkage on the Unique ID column.

Just another day as a researcher in a slightly less perfect world…

Crash Dataset				
	First	**Last**	**State**	**Crash Variables**
	Stephanie	Jansson	IA	…
	Divyanka	Pawaskar	NY	…
	Tara	Sangal	NY	…

Hospital dataset				
	First	**Last**	**State**	**Hospital Variables**
	Stephanie	Jansson	IA	…
	Tara	Sangal	NY	…
	Divyanka	Pawaskar	NY	…

Linking these is still pretty easy! Just do deterministic linkage using the first and last name.

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In the real world, we struggle.

Crash Dataset				
	First	Last	State	Crash Variables
	Shepanie	Not available. Sorry!	ID	…
	Divyanka		NY	…
	Tara		<N/A>	…

???

Hospital dataset				
	First	Last	State	Hospital Variables
	Stephanie	Jansson	IA	…
	Tara	Sangal	NY	…
	Divyanka	Pawaskar	NY	…

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DETERMINISTIC VS. PROBABILISTIC

A ***rules-based*** approached for joining records together--similar to "merging" two datasets together using identifiers

Ex: Crash Record X either matches Hospital Record Y, or it doesn't

An ***evidence-based*** approach for joining records together--yields a "match probability" for every potential pair of records

Ex: Crash Record X matches Hospital Record Y with probability 0.865

Does **not guarantee that all links are true matches**

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OUR HERO: THE FELLEGI-SUNTER MODEL

- The Fellegi-Sunter model is the statistical model that underlies Splink's methodology to perform probabilistic linkage
- It can compare records across multiple fields (Name, DOB, zipcode, etc.)
- It produces match probabilities that help us determine whether two records might correspond to the same entity

FELLEGI-SUNTER MODEL: THREE MAIN PARAMETERS

LAMBDA (λ)

- Probability that any two records match
- $\lambda = \Pr(Records\ match)$
- A measure of how likely a match is without information from data (prior probability)

M-PROBABILITY

- Probability of observing a record pair **given** the two records truly match
- $m = \Pr(Observation | Records\ match)$
- A measure of data reliability: how often might a person's DOB information change legitimately through data error?

U-PROBABILITY

- Probability of observing a record pair **given** the two records ***don't match***
- $\mathrm{u} = \Pr(Observation | Records\ do\ not\ match)$
- A measure of coincidence/cardinality: How many people might share the same DOB?

WHAT IS A "MATCH WEIGHT"?

- A **match weight** is a number that measures the likelihood of two records belonging to the **same entity**.
- Higher weight → more likely the records match.
- Lower weight → less likely the records match.
- Combines evidence from **different fields** (e.g., name, address, date of birth).

2	+	-2	+	5	=	5
Sex (matching)		DOB (not matching)		Last name (matching)		

Partial match weights

Total match weight

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WHAT GOES INTO A MATCH WEIGHT

- **Field comparisons:** Compare each column (e.g.,first name, last name, email).

LAMBDA (λ)

- We select a **match probability threshold** to classify matches and non-matches.
  - Ex: Only retaining matches where the match probability is 0.8 or higher

M-PROBABILITY

- **Agreement probabilities (m- values) :** How often values match when records are truly the same.

U-PROBABILITY

- **Disagreement probabilities (u- values):** How often values match by **chance** when records are different.

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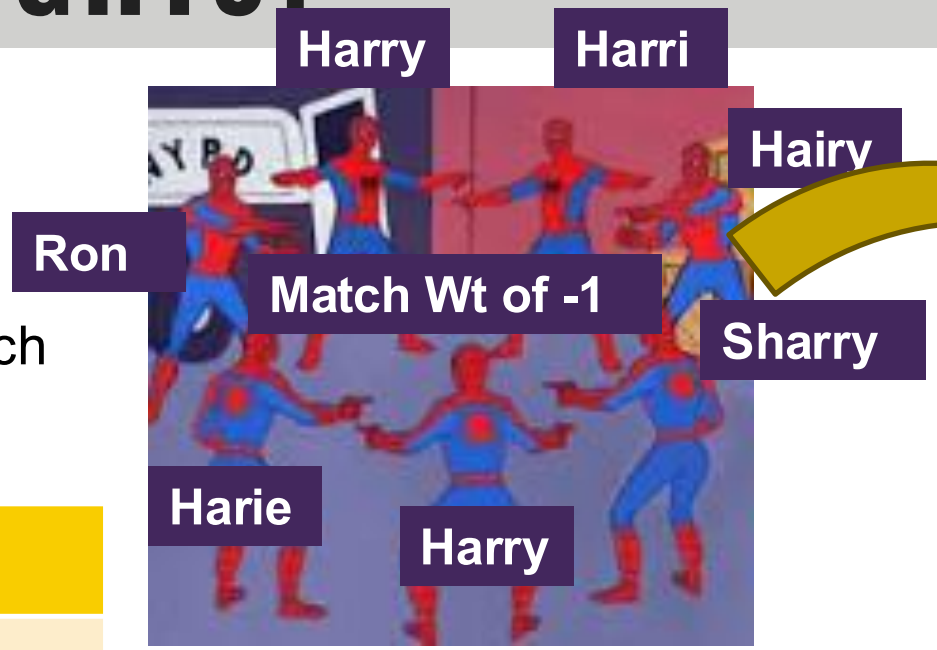
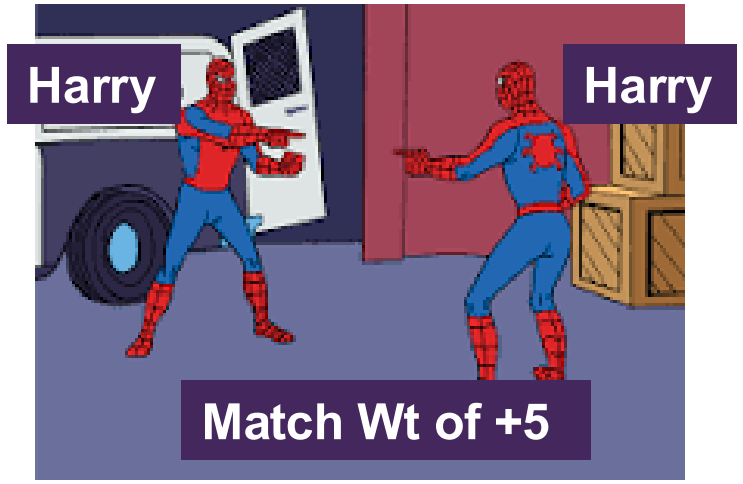
WHAT ARE THESE WEIGHTS?

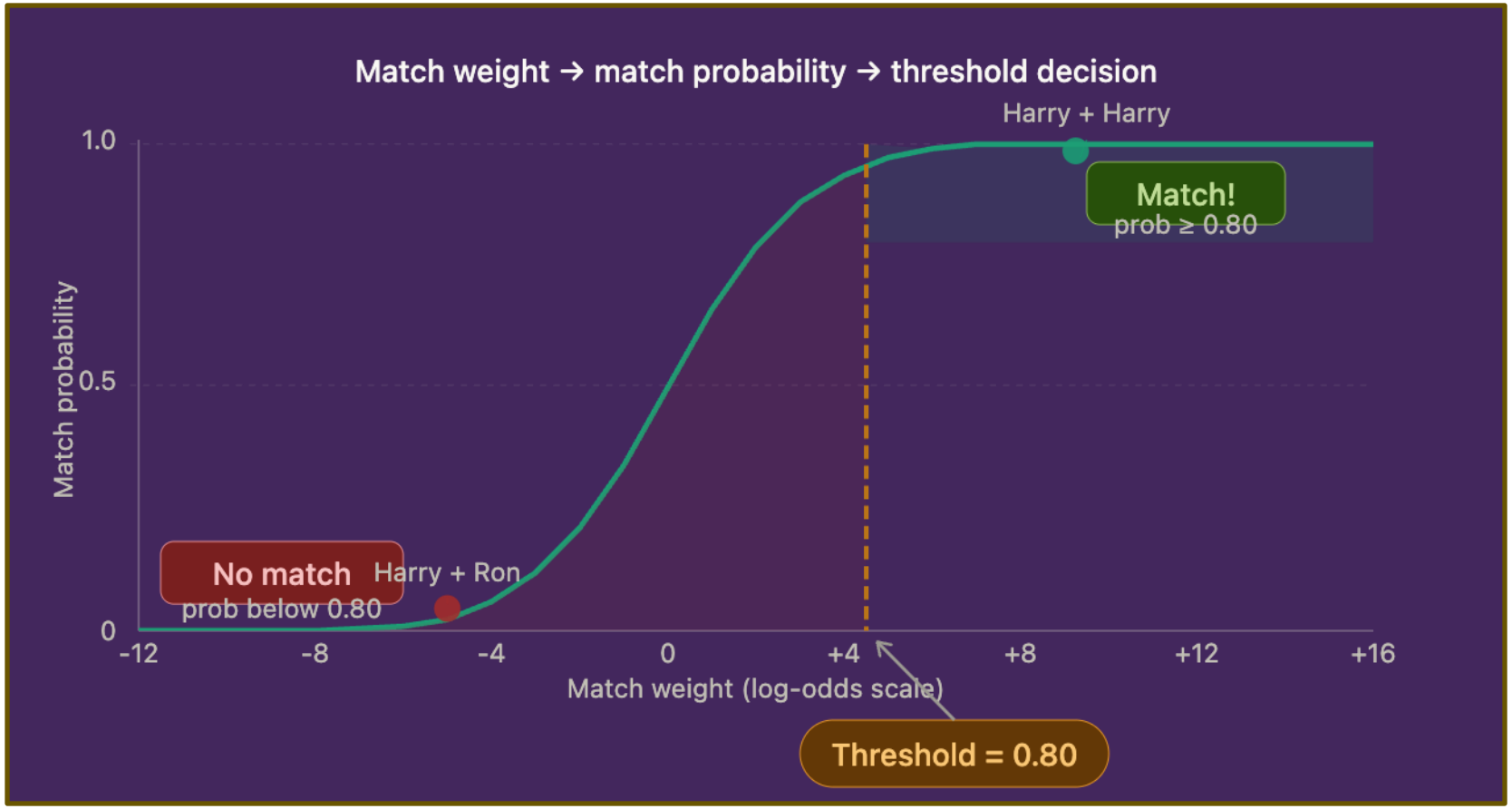
- **Weight calculation:**
  - Positive weight = evidence of a match
  - Negative weight = evidence of a non-match



Field	Record A	Record B	Result	Match Weight
First Name	Harry	Harrie	Strong Similarity	+3
Last Name	Potter	Potter	Exact Match	+5
Age	15	40	No Match	-4



HOW MATCH PROBABILITIES ARE CALCULATED



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A QUESTION FOR THE AUDIENCE:

What do you think is a good choice of match probability threshold?

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ACTIVITY: FINDING MATCHES IN A SMALL DATASET

ACTIVITY: FINDING MATCHES IN A SMALL DATASET

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Submit your answers here!

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PART 2: SOFTWARE DEMONSTRATION WITH SPLINK

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PART 2: SOFTWARE

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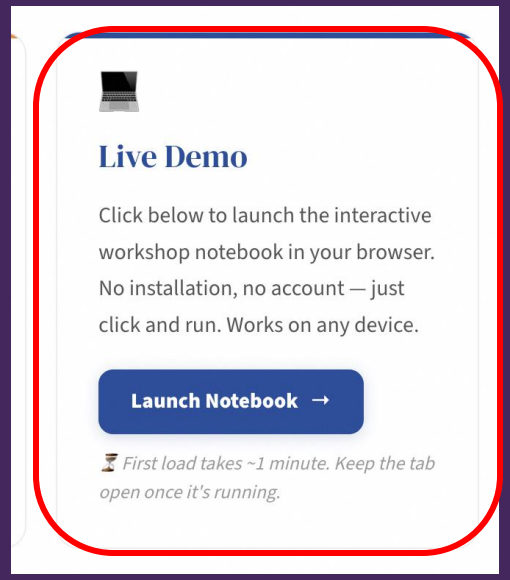
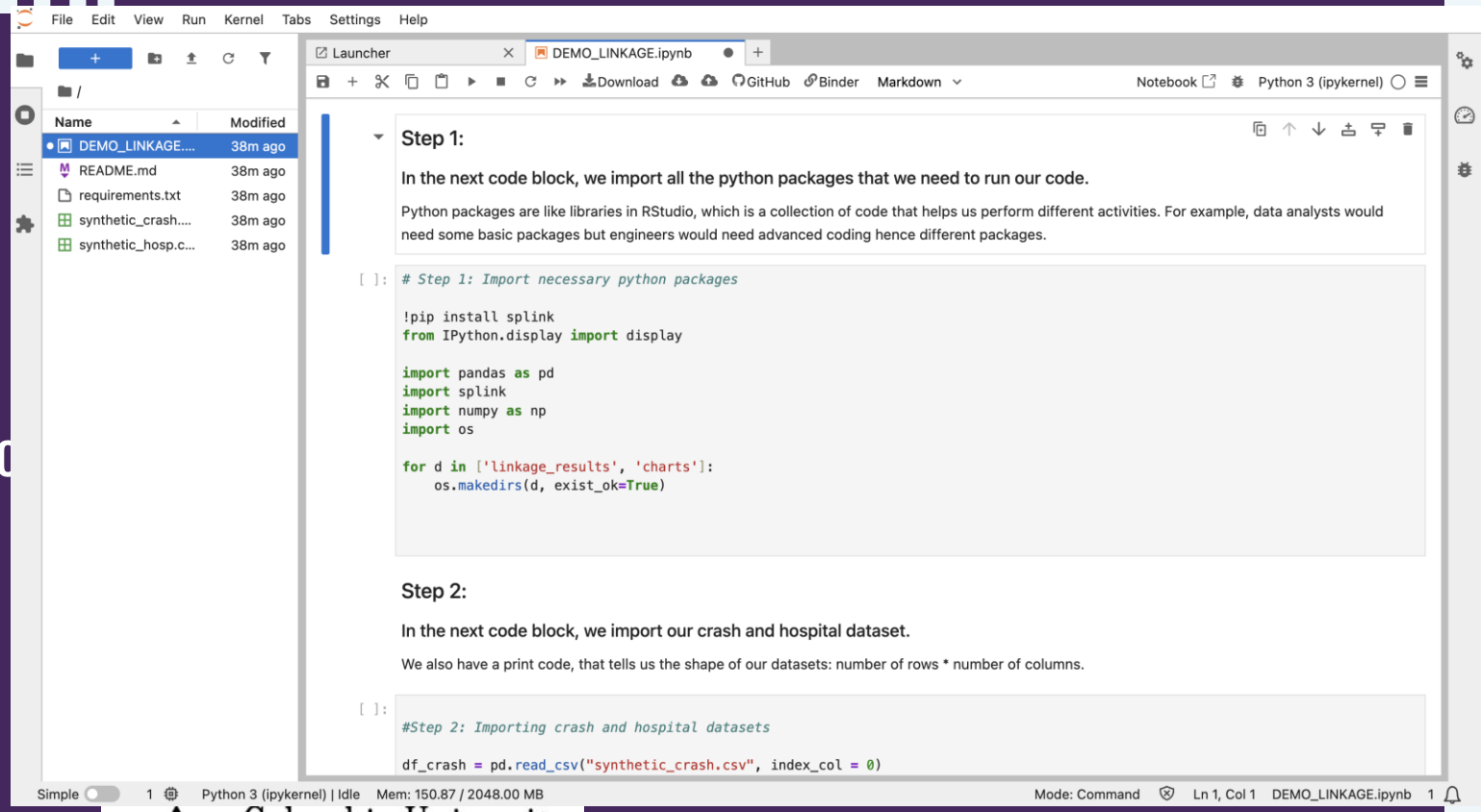
DEMONSTRATION WITH SPLINK



Splink Demo



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WHAT IS SPLINK AND WHY DO WE USE IT?

- **Splink** is a Python package that **implements the Fellegi-Sunter model** for probabilistic record linkage

SPLINK
Data linkage at scale

- It can de-duplicate and link datasets lacking unique identifiers

Fast

Accurate

Scalable

Interactive visualizations

Abundant online resources

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SOFTWARE CONSIDERATIONS

- Splink is a Python software package, requiring a Python installation
- We have explored options for accessing Splink, depending on coding environment preferences:
  - Run Python directly in any python-compatible IDE or notebook

    jupyter Visual Studio Code Google Colab
  - Run Python in code chunks in RStudio using `reticulate` package

    RStudio R

TODAY'S DEMO DATA: A SYNTHETIC NEW YORK STATE CODES LINKAGE

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FEATURES OF OUR SYNTHETIC DATA

- 1,000 unique crash records for 667 individuals; 1,000 hospital records for 667 individuals; **333 true matching record pairs**
- Some crash records don’t link to hospital records, and vice versa (i.e. some crashes never get reported)
- Missing values: higher data missingness for vehicle passengers and non-motorists (pedestrians, cyclists)
- Crash dates and hospital admission dates don’t line up (people may go to the hospital one or more days after a crash)
- Location info (county, zip code) often mismatches even for true links

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DEMO: DATA CLEANING AND STANDARDIZATION

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DATA CLEANING AND STANDARDIZATION

Ensuring compatibility:

- Rename columns to match between datasets
- Convert all data types to strings or numeric and ensure consistency between datasets
- Recode variables so that values match between datasets

Standardizing variables:

- Convert strings to all lowercase, remove special characters
- Standardize date formats
- Ensure NAs and NULLs are entered as such (some data sources use “77777” or “99999”)

First Name	Last Name	Dob	House
Harry	Potter	31-07-80	Gryffindor
Ginny	Weasley!	999	Gryffindor
Hermione	Granger	NULL	Gryffindor

➡

First Name	Last Name	Dob	House
harry	potter	1980-07-31	gryffindor
ginny	weasley	NULL	gryffindor
hermione	granger	NULL	gryffindor

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DEMO: EXPLORATORY ANALYSIS

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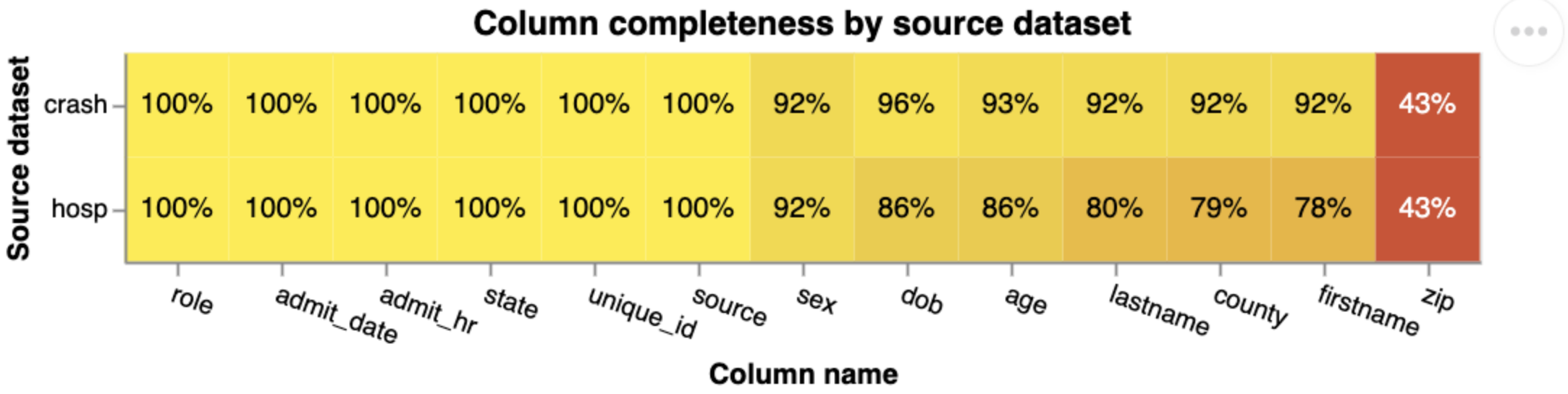
EXPLORATORY ANALYSIS: DATA MISSINGNESS

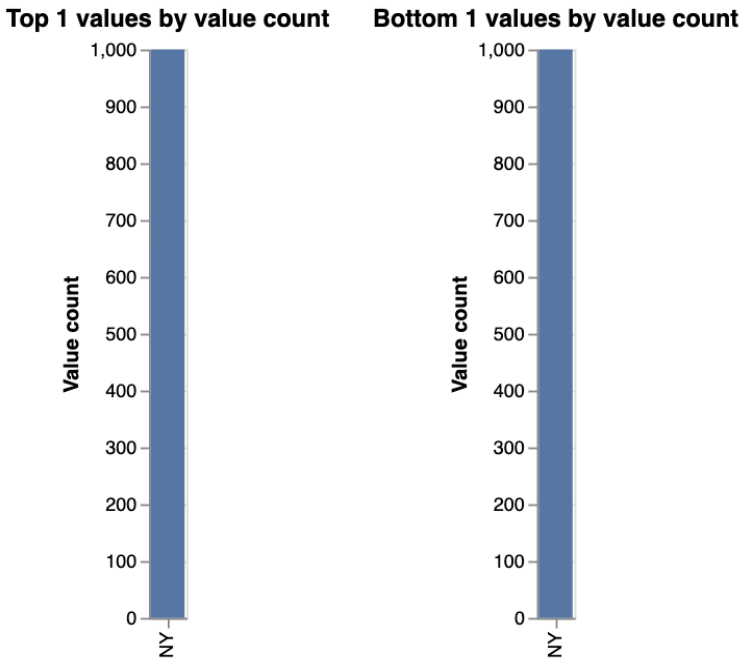
- Examine each variable individually (DOB, sex, zipcode, etc.) and check for missing data.
- Variables with high missingness can still be used for linkage, but usefulness may vary.



EXPLORATORY ANALYSIS: DISTRIBUTIONS OF VARIABLES

Pay attention to the distributions of distinct values for each variable.

Distribution of variable: State



Interpretation:

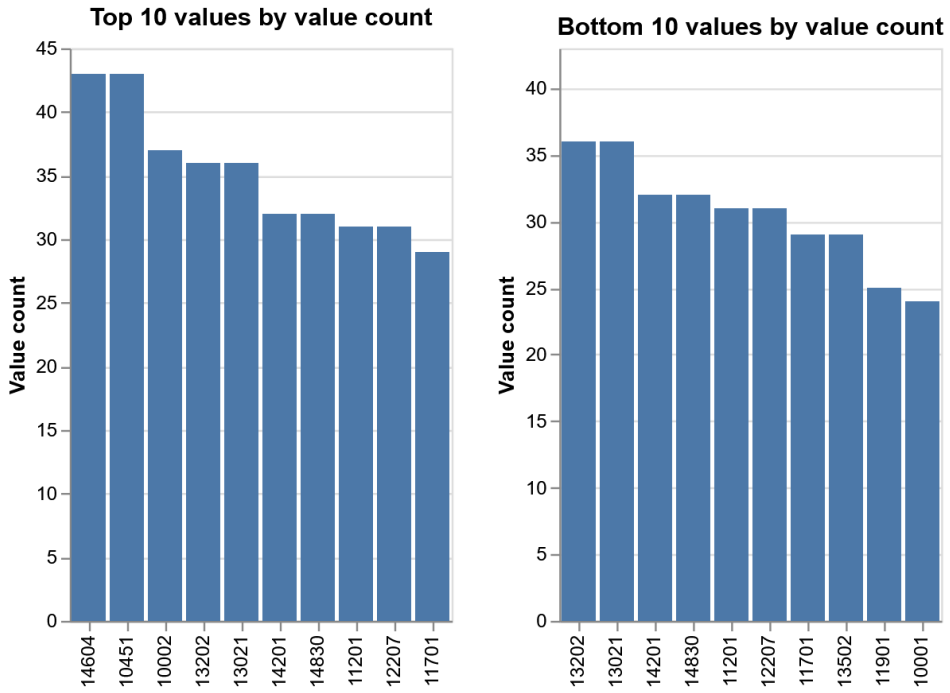
- The only value observed of the state variable is “NY”.
- This won’t help us distinguish between individual people, so we shouldn’t include it in the linkage

https://moj-analytical-services.github.io/splink/charts/profile_columns.html

EXPLORATORY ANALYSIS: DISTRIBUTIONS OF VARIABLES

Pay attention to the distributions of distinct values for each variable.

Distribution of variable: Zip Code



Interpretation:

- Zip code is distributed over a wide range of values.
- It can be helpful in distinguishing individuals.
- Consider this in conjunction with missingness.

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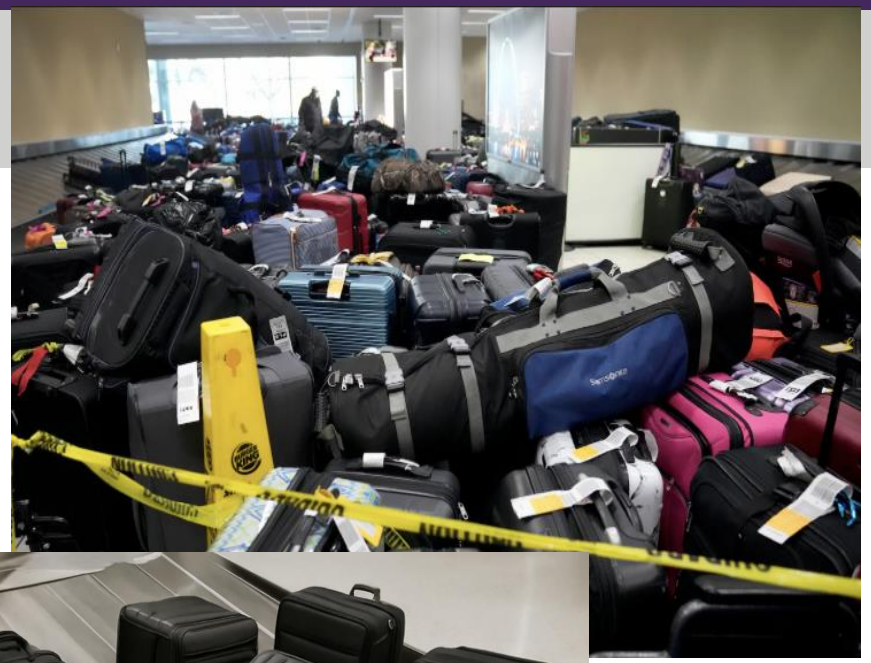
https://moj-analytical-services.github.io/splink/charts/profile_columns.html

DEMO: BLOCKING RULES



1. Black bag
2. Delta Airlines
3. Other bag features
4. Baggage claim tag no.
5. Contents of the bag

BLOCKING RULES

First Name	Last Name	Age	Sex
Ginny	Weasley	36	Female
Harry	Potter	45	Male
Harry	Potter	40	Male

×

First Name	Last Name	Age	Sex
Ginny	Weasley	36	Female
Harry	Potter	45	Male
Harry	Potter	40	Male

= 3*3
= 9

In practice, linkage datasets can be large:
For example, every year New York State links around 883,000 crash records to 156,500 emergency department records.

883,000* 156,500 ~ 138 billion possible combinations

- Computationally intensive
- Time-consuming to process
- High memory and storage requirements

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firstname	lastname	sex	age	dob	role	crash date	crash hour	county
Alice	Smith	M	28	1996-05-10	Driver	3/1/2024	14	Kings
Bob	Smith	F	45	1979-08-12	Passenger	3/2/2024	9	Queens
Carlos	Garcia		33	1991-01-20	Driver	3/3/2024	16	Bronx
	Patel	F	52	1972-11-03	Driver			New York
Eric	Chen	M			Passenger	3/5/2024	20	Westchester
Fiona	Taylar	F	61	1963-02-28	Driver	3/6/2024	8	Suffolk
Frank	Miller	M	41	1983-07-14	Passenger	3/7/2024	12	Erie
Alice	Miller	F	41	1983-09-22	Driver	3/7/2024	15	
Henry	Ward	M	62	1962-04-01		3/8/2024	18	Monroe
Ivy	Fox	F	19	2005-11-30	Passenger	3/9/2024		Onondaga

firstname	lastname	sex	age	dob	role	admit date	admit hour	county
Alex	Smith	M	28	1996-05-10		3/2/2024	15	Kings
Bob	Smith	F	45	1979-08-12	Passenger	3/3/2024	10	Queens
Karlos	Garcia	M	33	1991-01-20	Unknown	3/7/2024	17	Bronx
Diana	Patel	F	52	1972-11-03	Driver	3/5/2024	12	Bronx
Eric	Chen	M	39	1985-04-15	Passenger	3/6/2024	21	Westchester
Fiona	Taylor	F	60	1963-02-28	Driver	3/7/2024	9	Suffolk
Franklin	Miller	M	41	1983-07-14	Passenger	3/8/2024	13	Erie
Alice		F	41	1983-09-22		3/8/2024	16	Erie
Henri	Ward	M	62	1962-04-01	Driver	3/9/2024	19	Monroe
Alice	Smith	F	19	2005-11-30	Passenger	3/10/2024	22	Kings

DOB
gender
Age
Lastname
Crash date
Firstname
DOB
Age
Admit date
Lastname
gender
Firstname
YOU/SPLINK

firstname	lastname	sex	age	dob	role	crash date	crash hour	county
Alice	Smith	M	28	1996-05-10	Driver	3/1/2024	14	Kings
Bob	Smith	F	45	1979-08-12	Passenger	3/2/2024	9	Queens
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firstname	lastname	sex	age	dob	role	admit date	admit hour	county
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Henri	Ward	M	62	1962-04-01	Driver	3/9/2024	19	Monroe
Alice	Smith	F	19	2005-11-30	Passenger	3/10/2024	22	Kings

firstname	lastname	sex	age	dob	role	crash date	crash hour	county
Alice	Smith	M	28	1996-05-10	Driver	3/1/2024	14	Kings

firstname	lastname	sex	age	dob	role	admit date	admit hour	county
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Alice	Smith	F	19	2005-11-30	Passenger	3/10/2024	22	Kings

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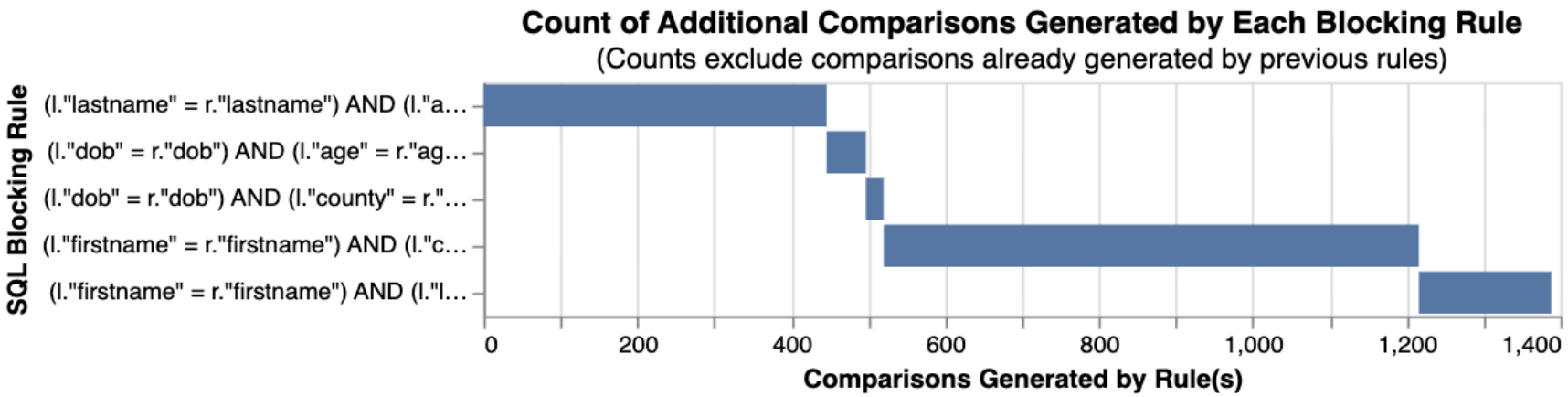
BLOCKING RULES APPLICATION IN SPLINK

1. Cumulative Comparisons Chart
2. Pre-training model parameters
3. EM to estimate model parameters



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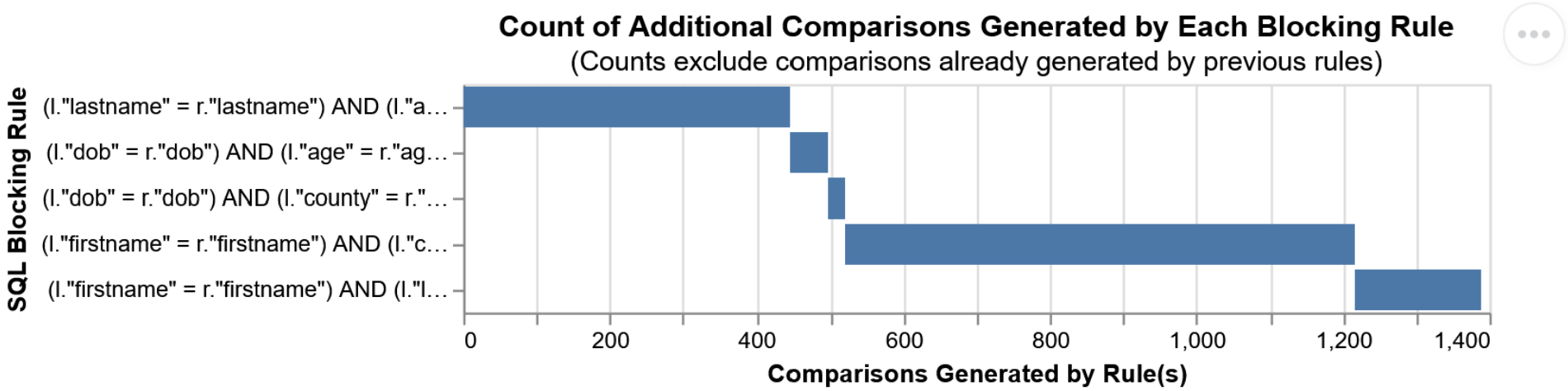
CUMULATIVE COMPARISONS CHART

Cumulative Comparisons Chart



Interpretation:

- This chart shows how many comparisons each blocking rule contributes to the linkage
- The 1st and 4th blocking rules contribute the most record comparisons
- We create blocking rules to limit the number of comparisons made (to save on computation)
- **Without blocking rules**, we would have 1,000,000 total record comparisons
- **With blocking rules**, we have 1,400 total record comparisons

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DEMO: COMPARISONS

COMPARISONS AND COMPARISON LEVELS

- Applied only to candidate pairs created after blocking
- **Comparisons** define which fields are compared (e.g., Age vs Age)
- **Comparison levels** define how similarity in columns is assessed (e.g., Exact Match)
- Splink has various built-in comparisons:

JaroWinkler Comparison	Names, strings with typos
Levenshtein Comparison	Free-text fields, minor edits
DateOfBirth Comparison	Dates, birthdays with partial matches
Custom Comparison	Customize for any column

```
comparisons = [
    cl.ExactMatch("firstnamef2"),
    cl.ExactMatch("lastnamef2"),
    cl.ExactMatch("lastnamel2"),

    cl.ExactMatch("sex"),
    cl.ExactMatch("dob"),
    cl.ExactMatch("age"),

    cl.ExactMatch("county"),
    cl.ExactMatch("zip"),


    cl.ExactMatch("role"),
    cl.ExactMatch("collision"),
]
```

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PRE-TRAINING PARAMETER ESTIMATION

Estimate baseline λ and U parameter values using random sampling

Set deterministic rules to estimate probability that two random samples match

⇓

Use random samples to calculate baseline U values

```
deterministic_rules = [
    block_on("firstnamef2", "lastnamef2", "lastnamel2"),
    block_on("dob", "county", "zip"),
    block_on("lastnamef2", "lastnamel2", "age"),
]

linker_detailed.training.estimate_probability_two_random_records_match(
    deterministic_rules, recall=0.8
)
```

```
linker_detailed.training.estimate_u_using_random_sampling(max_pairs=1e6)
```

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DEMO: EM TRAINING

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EXPECTATION MAXIMIZATION (E-M) ALGORITHM

We use the EM algorithm to get parameter estimates for **M**, **U**, and λ, and thus our match weights!

CANDIDATE PAIRS

Generated via blocking rules

MATCH WEIGHTS

Calculated for all columns not used for blocking

→

E-STEP

The Expectation Step estimates match probability for each candidate pair using current M and U probabilities

→

M-STEP

The Maximization Step updates the M and U probabilities based on the E-step's output

→

Has the model converged? (*M and U probabilities stop changing significantly between iterations*)

NO → E-STEP

YES
Final match weights and match probabilities are calculated

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DEMO: VISUALIZATIONS AND MODEL EVALUATION

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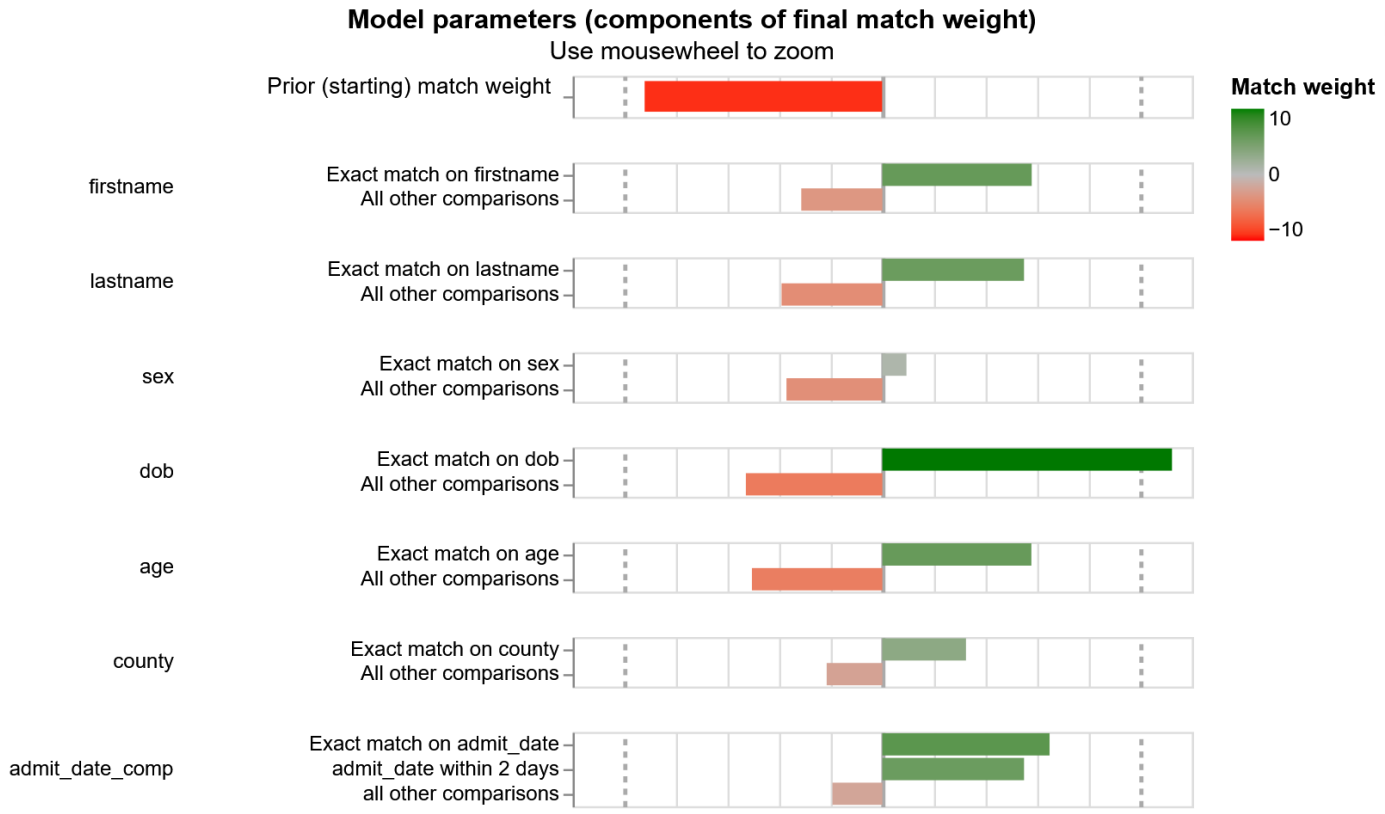
MATCH WEIGHTS PARAMETERS CHART



Green Bar: the comparison contributes positive weight to the final match weight

- Ex: DOB, when matching exactly, is extremely helpful in distinguishing records (large green bar)

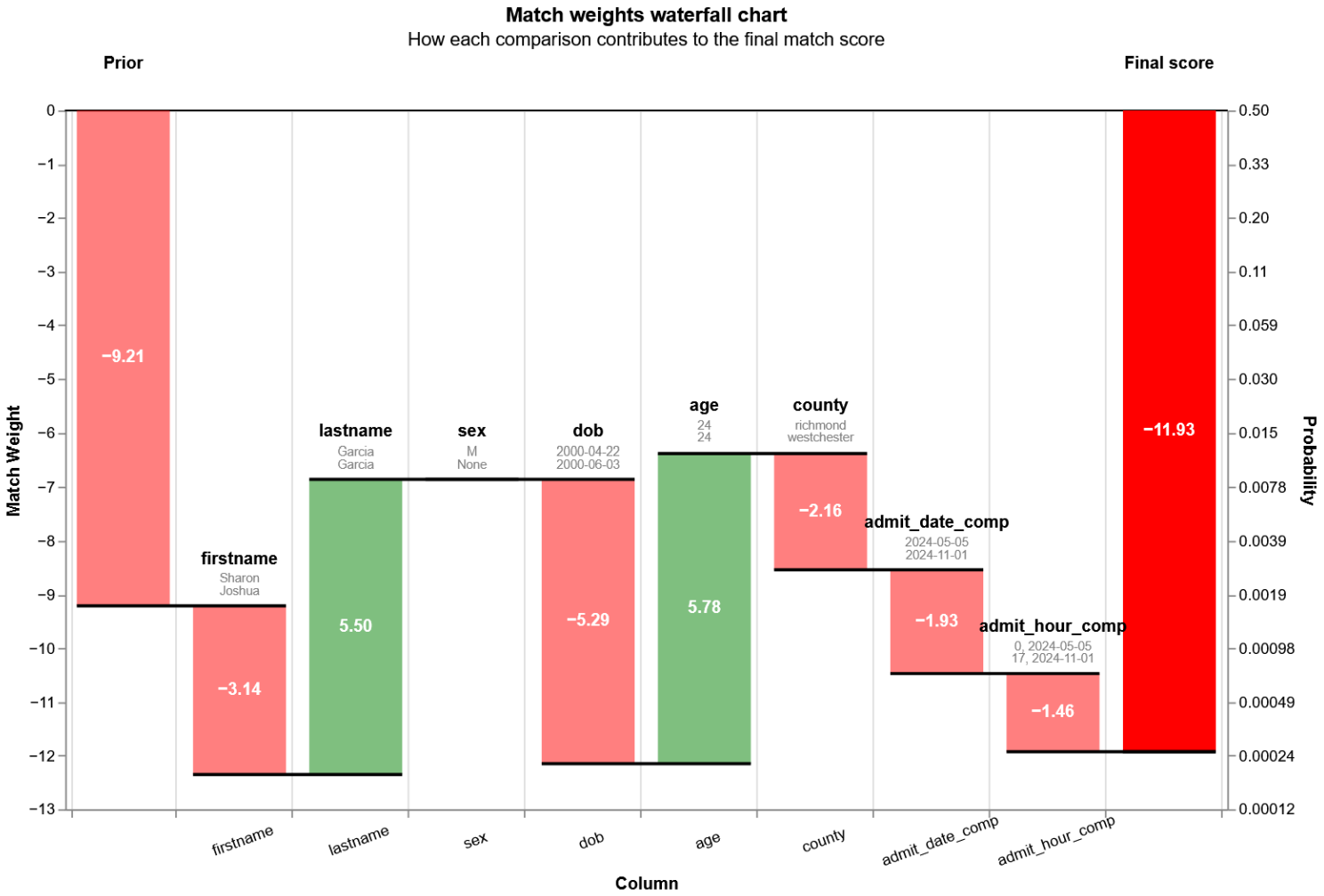
Red Bar: when not matching, the comparison contributes a negative weight to the final match weight

- Ex: County, when not matching, gives a smaller negative weight because it doesn't match a lot of the time, even for record pairs that match well on other variables

WATERFALL CHARTS

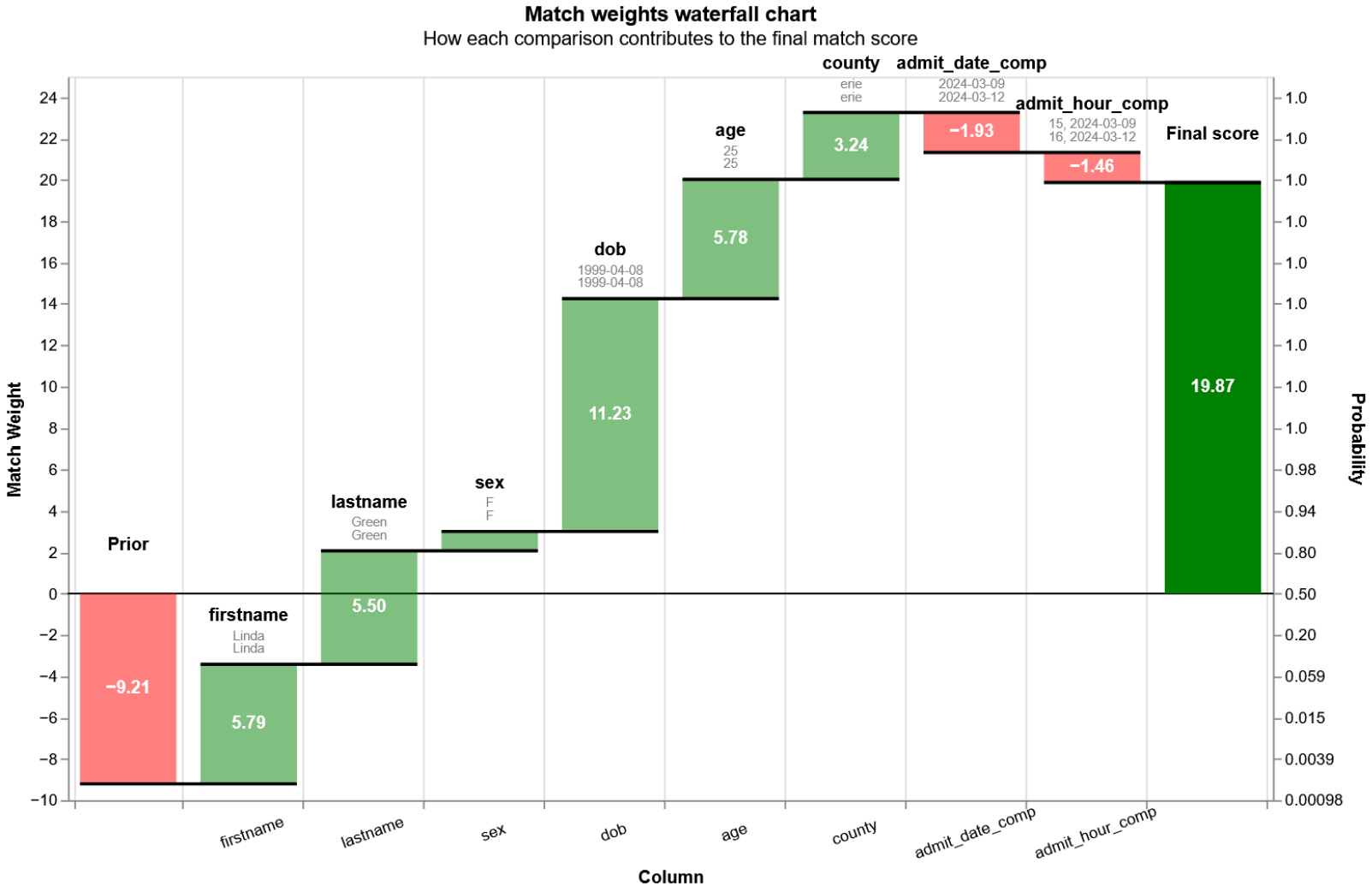
WATERFALL CHART #1

Example 1: This pair of records is ***probably NOT a match!***



WATERFALL CHART #2

Example 2: This pair of records is ***probably a match!***



CONSIDERATIONS FOR LINKING LARGE DATASETS

- **Data quality:** Missingness, inconsistent formats, and non-informative columns have a huge impact in large dataset linkage.
- **Strategic Blocking rules:** Use high-quality, discriminative columns with minimal missing data.
- **Comparison Levels:** create custom comparisons or use out-of-the-box comparisons beyond “Exact Match”
- **Computation:** It is easy to run out of RAM and crash your software
  - Link one year’s worth of data at a time
  - Use more specific blocking rules

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RESOURCES WE RECOMMEND

1. Splink Documentation (explanations, tutorials, and examples)
2. Robin Linacre's Blog (intuitive explanations, diving deeper into probabilistic linkage issues)
3. Our workshop Github (CODES-specific code files, linkage troubleshooting suggestions)

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ACKNOWLEDGEMENTS

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